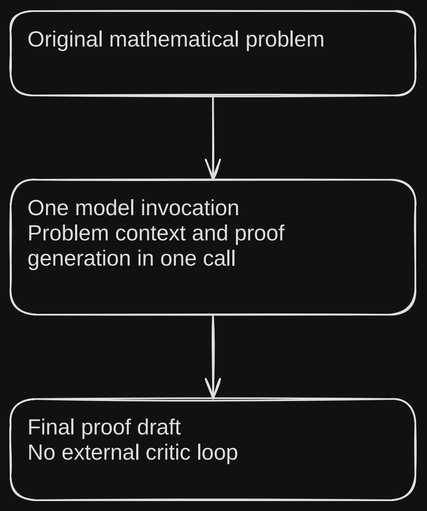
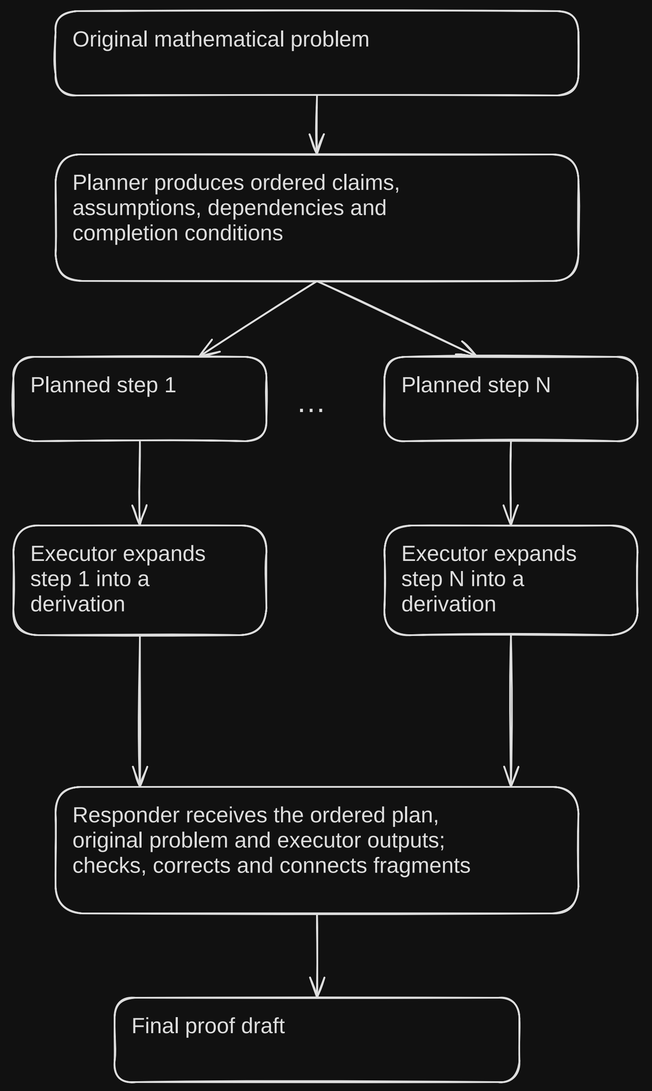
Same language model. Three ways to manage mathematical context.

Case study: “Prove the Prime Number Theorem” | IFM/K2-Horizon-375B-A23B

1. Standalone model



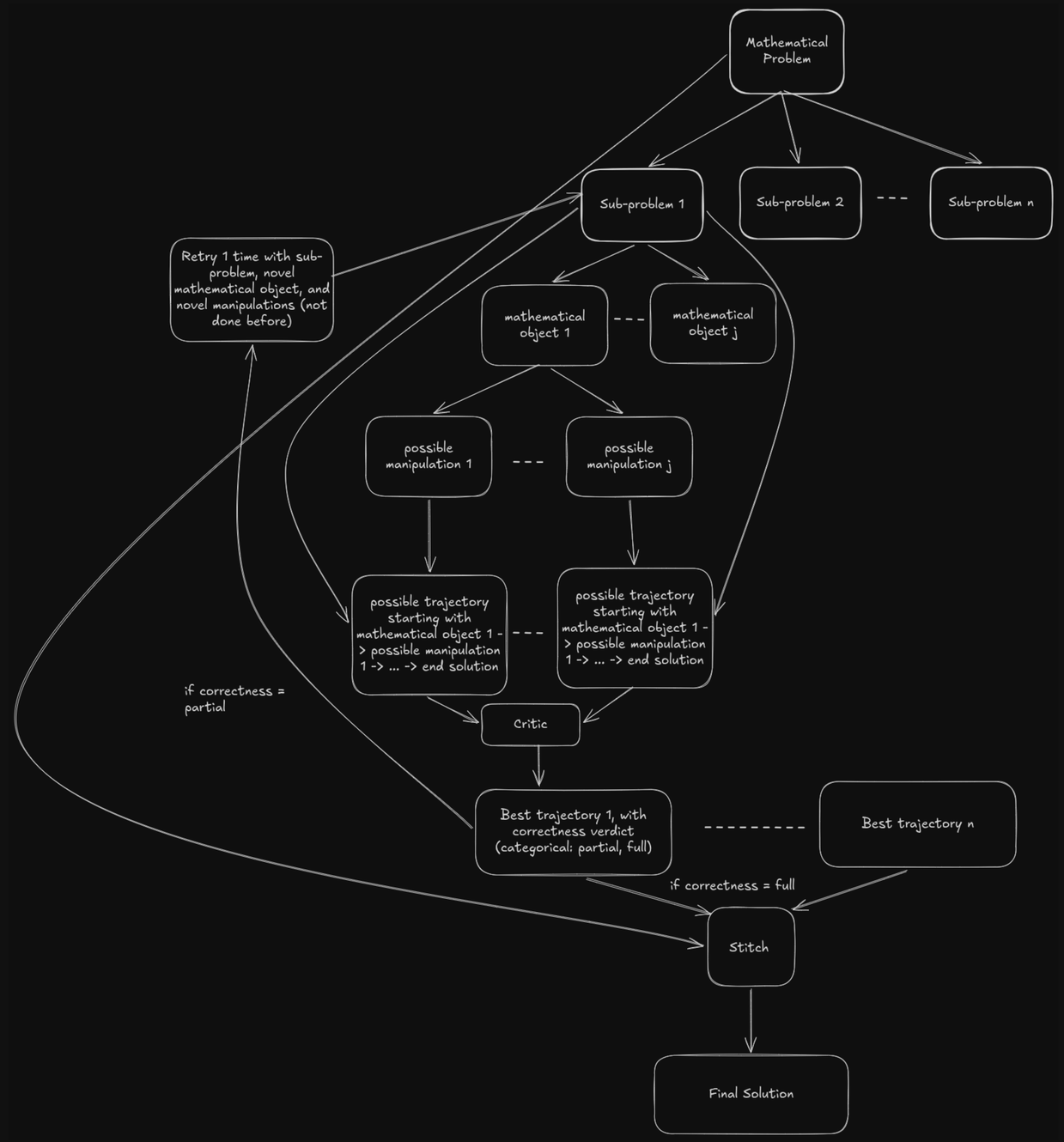
2. Planner–executor



No automatic replanning or critic loop.
Blocked dependencies remain partial.
Step artifacts are retained in SQLite.

Sequential execution in dependency order.
Each call receives the original problem, local claim and required predecessor proofs only.

3. Math Harness



Supplied source figure retained above. Current code: subproblems 1…i; objects 1…j; manipulations 1…k. All calls are sequential. Stitching also accepts partial or unranked work; a full verdict is not required.

OBSERVED OUTPUTS — qualitative review of one draft per approach

Standalone: invalid asymptotic step.
The PNT conclusion was not established.

Planner–executor: strongest correctness baseline in the supplied review; cites Wiener–Ikehara.

Math Harness: broader supporting derivations; local errors and an unproved Tauberian lemma.

Different call and token budgets; incomplete harness search. Model-assisted review, not a controlled benchmark or formal verification.

github.com/antibloch/math_harness